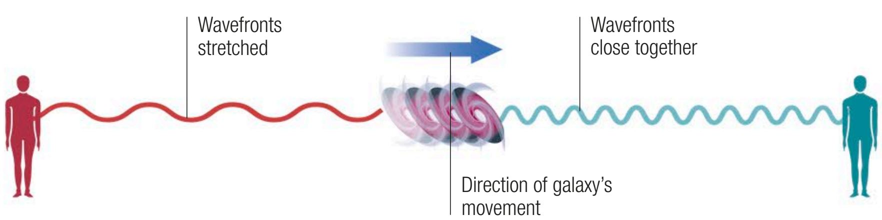
THE EXPANDING UNIVERSE

AS A GENERAL RULE, THE FARTHER A GALAXY IS FROM EARTH, THE FASTER IT IS MOVING AWAY FROM US. THIS IS VITAL EVIDENCE THAT THE UNIVERSE AS A WHOLE IS STILL EXPANDING. AND WHAT'S MORE, THAT EXPANSION IS ACCELERATING.

The key evidence for the expansion of the cosmos comes from the Doppler effect (a way of measuring the speed at which any light-emitting object is moving toward or away from Earth). This reveals that more distant galaxies are moving away from Earth more rapidly—a discovery best explained by the idea that space as a whole is expanding and carrying galaxies away from one another—rather like currants in a cake moving apart as the batter rises in the oven.

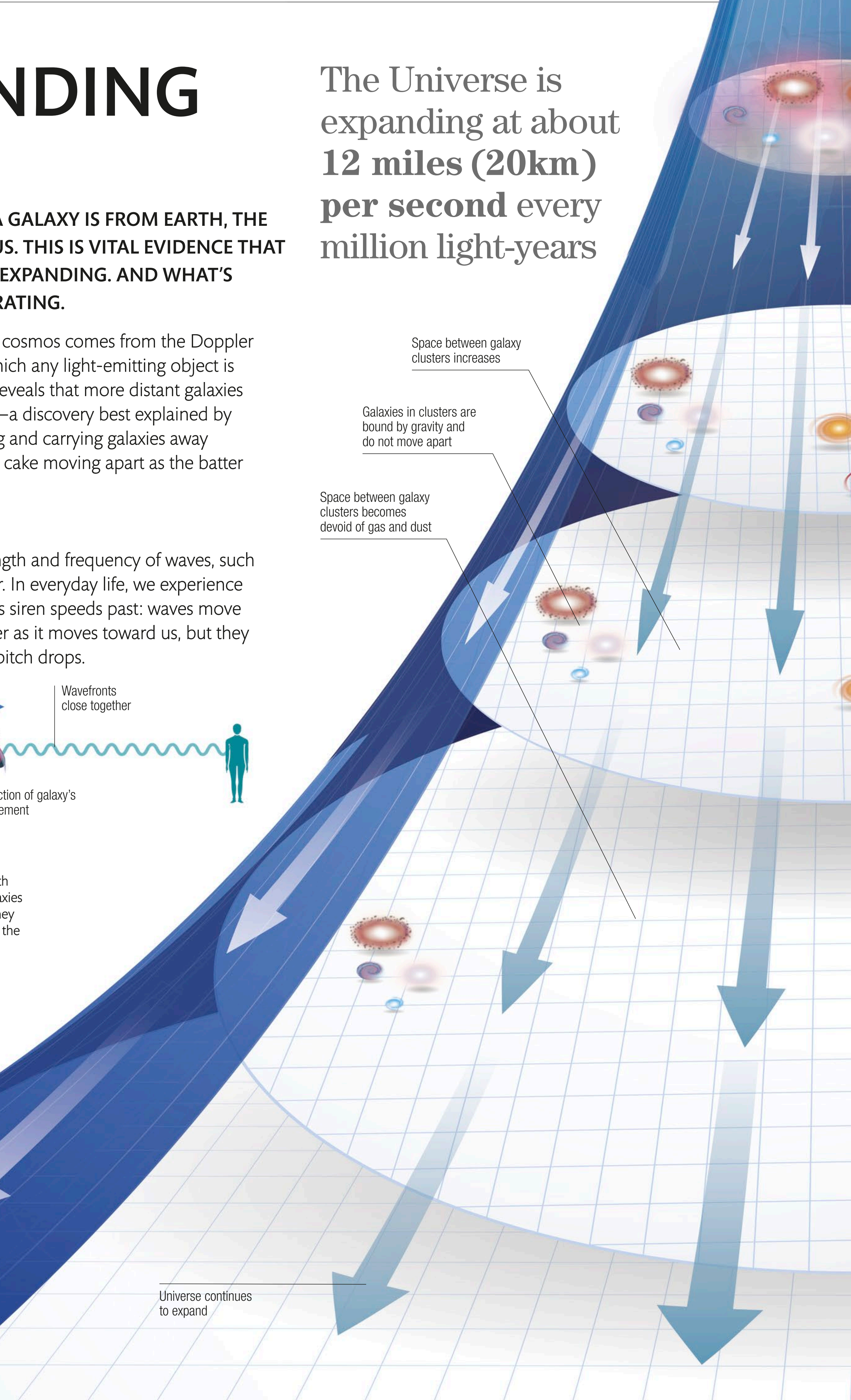
The Doppler effect and redshift

The Doppler effect is a shift in the wavelength and frequency of waves, such as sound or light, moving past an observer. In everyday life, we experience Doppler shift when an emergency vehicle's siren speeds past: waves move past us more rapidly and the pitch is higher as it moves toward us, but they reach us more slowly as it retreats, so the pitch drops.



△ **Redshift and blueshift**
Light from galaxies moving away from us has its wavelength stretched and therefore appears redder. When nearby galaxies move toward us, their wavelengths are compressed and they appear bluer. These shifts can be precisely measured from the way they affect spectral lines in a galaxy's light.

The Universe is expanding at about **12 miles (20km) per second** every million light-years



▷ **Cosmic expansion**
The expansion of the Universe is not simply a case of galaxies moving apart in space—most of it is caused by space itself expanding. This is a result of the Big Bang explosion that created not only matter but space and time themselves.